

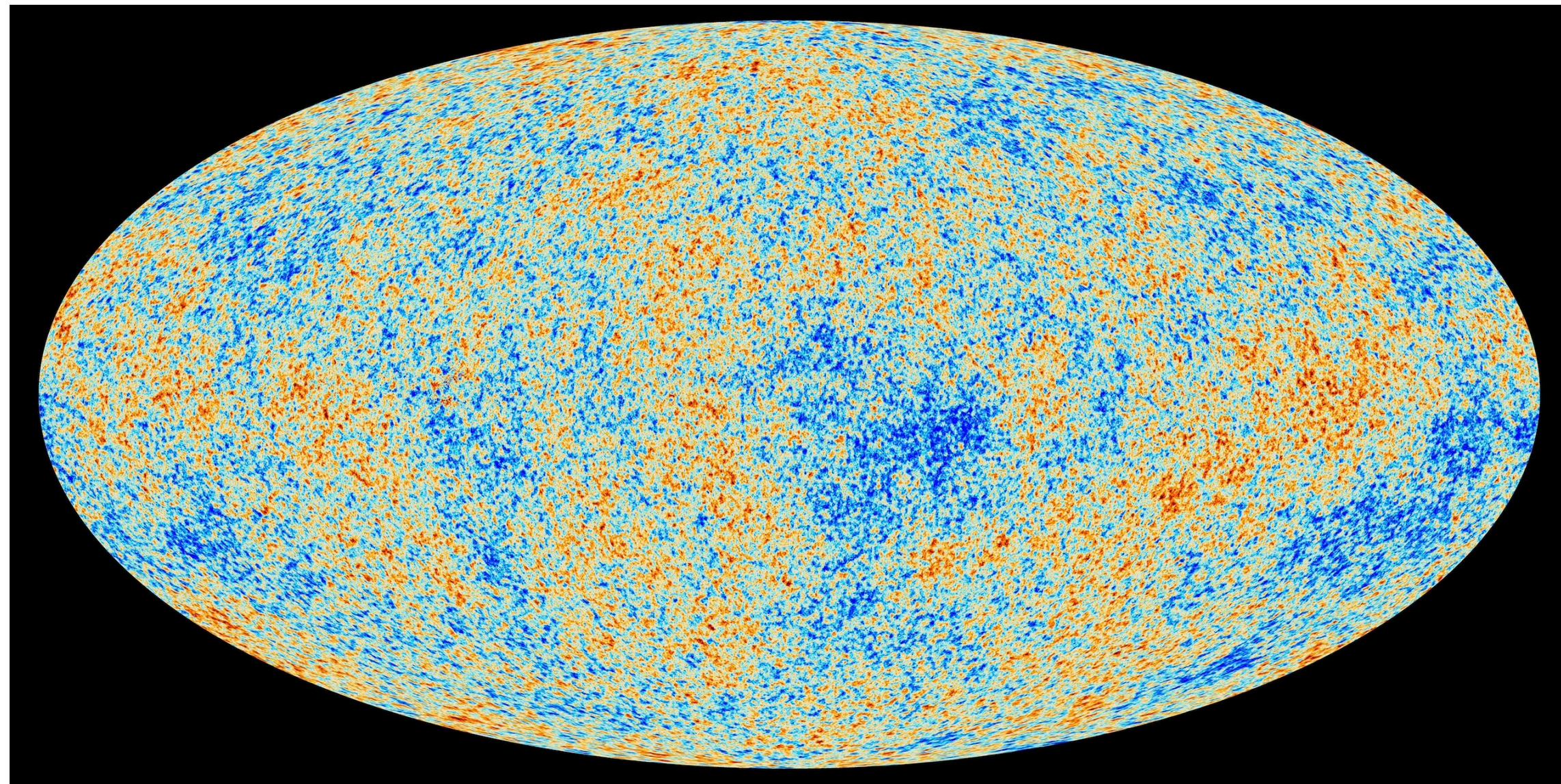
Can the Radiation Era Extend Back to the Bang?

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with Latham Boyle and Neil Turok

Higgs Centre for Theoretical Physics
University of Edinburgh

Seminar
Kobayashi-Maskawa Institute, Nagoya
6th July 2026

The Cosmic Microwave Background



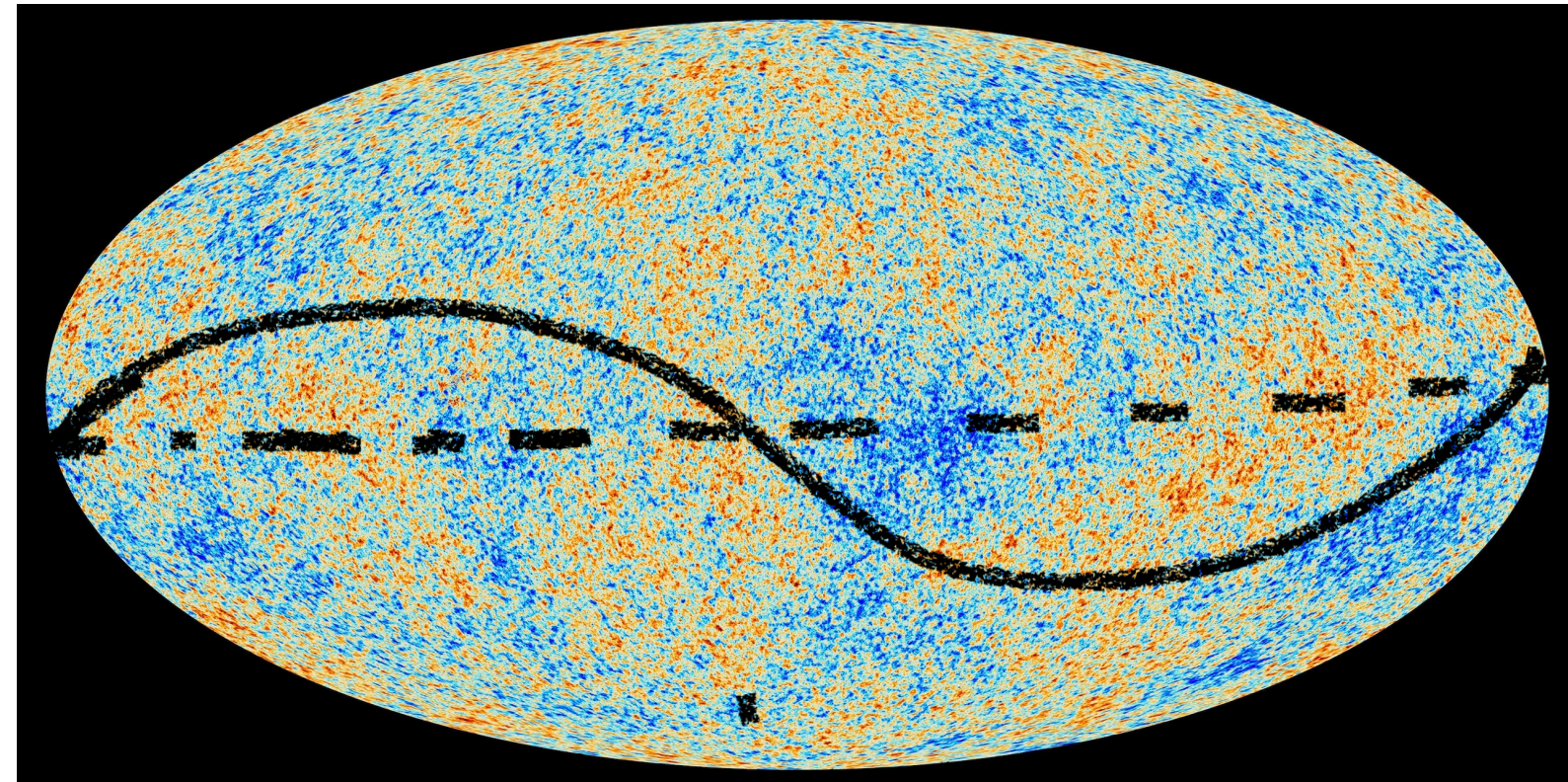
- • Homogenous
 - Isotropic
 - Spatially flat
- } (almost!)

Temperature Fluctuations!

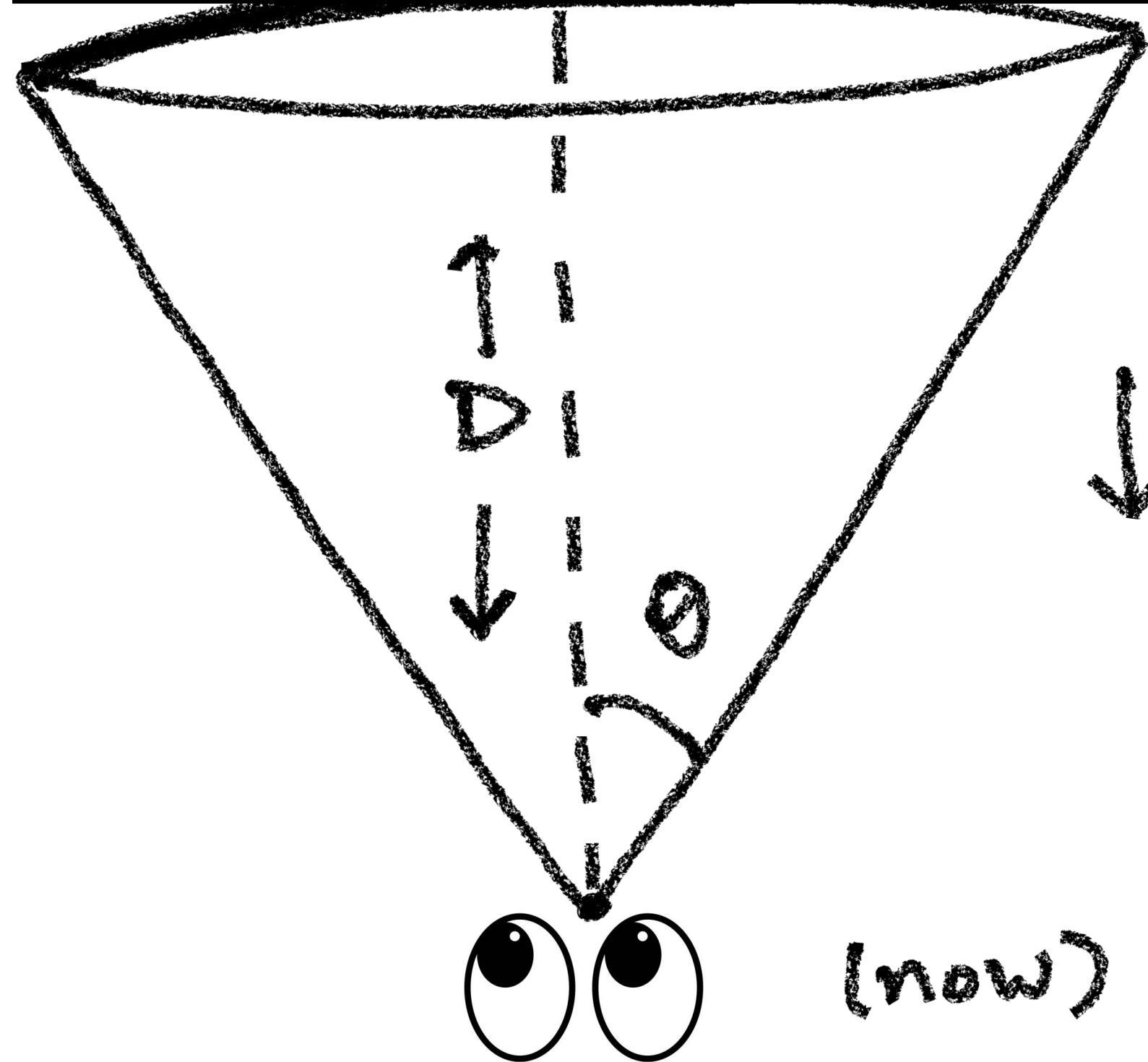
The Cosmic Microwave Background



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(last scattering)

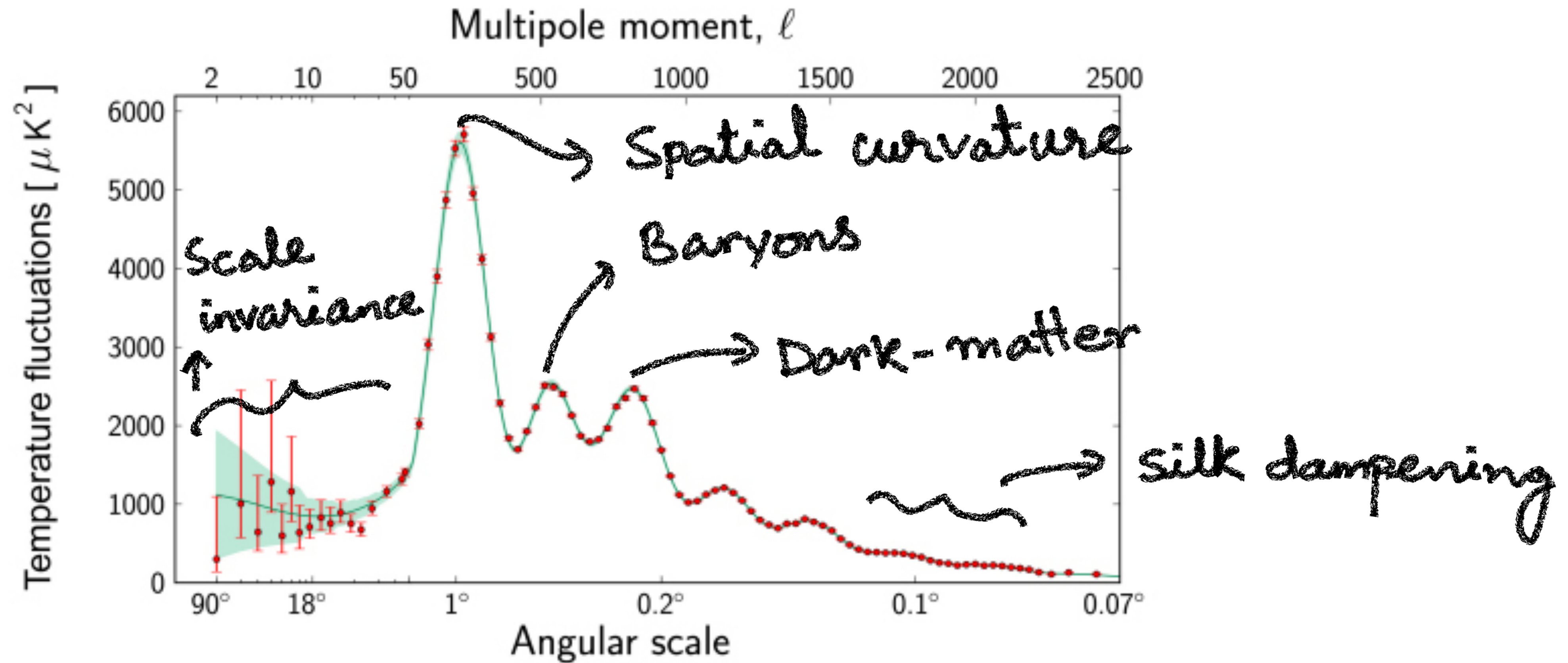


$$\frac{\lambda}{2} = \frac{\pi}{|k|} \approx D\theta = \frac{D\pi}{\ell}$$

$$|k| = \frac{\ell}{D}$$

The Cosmic Microwave Background

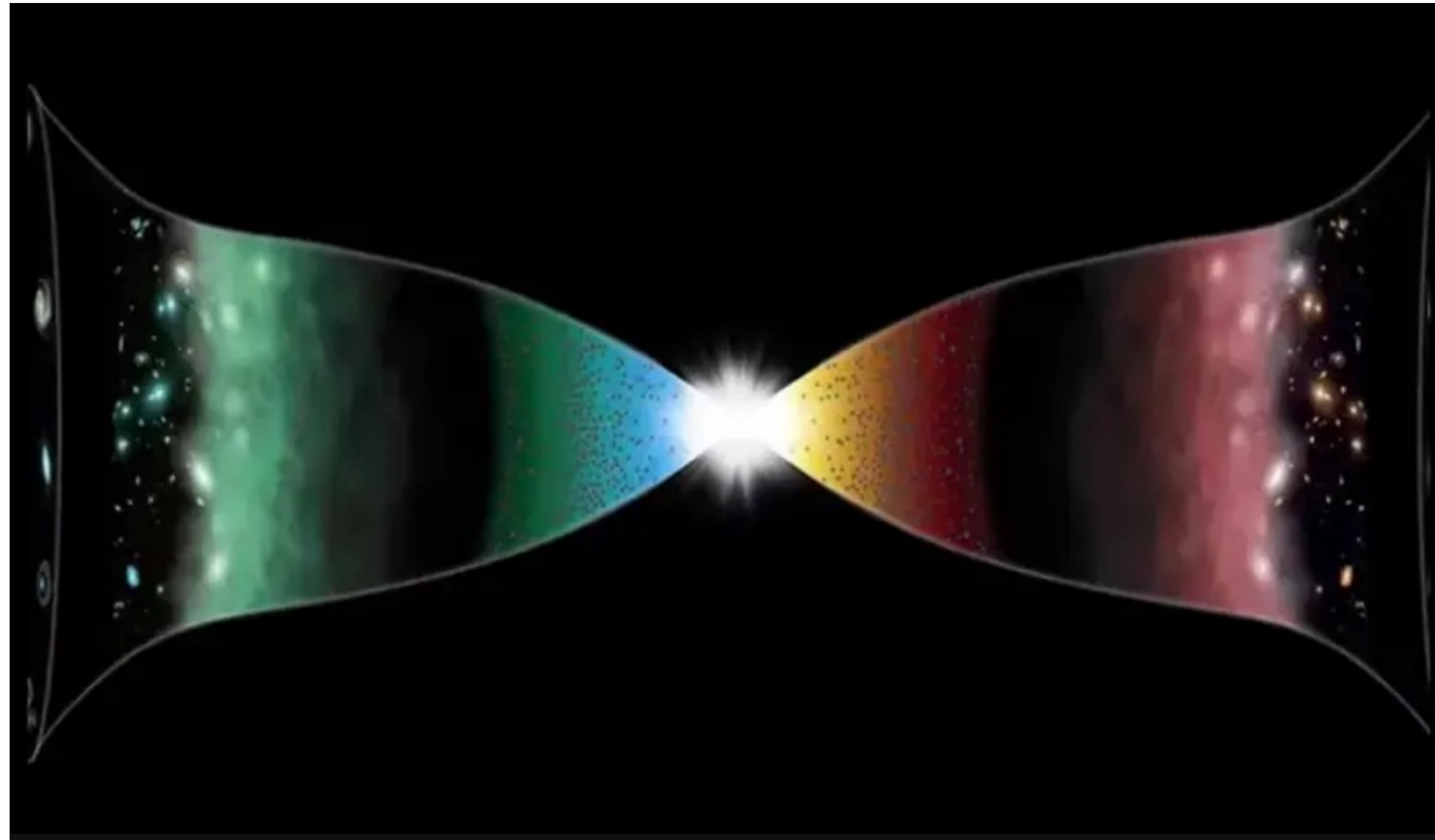
- Adiabatic
- Gaussian
- No tensor/vector perturbation



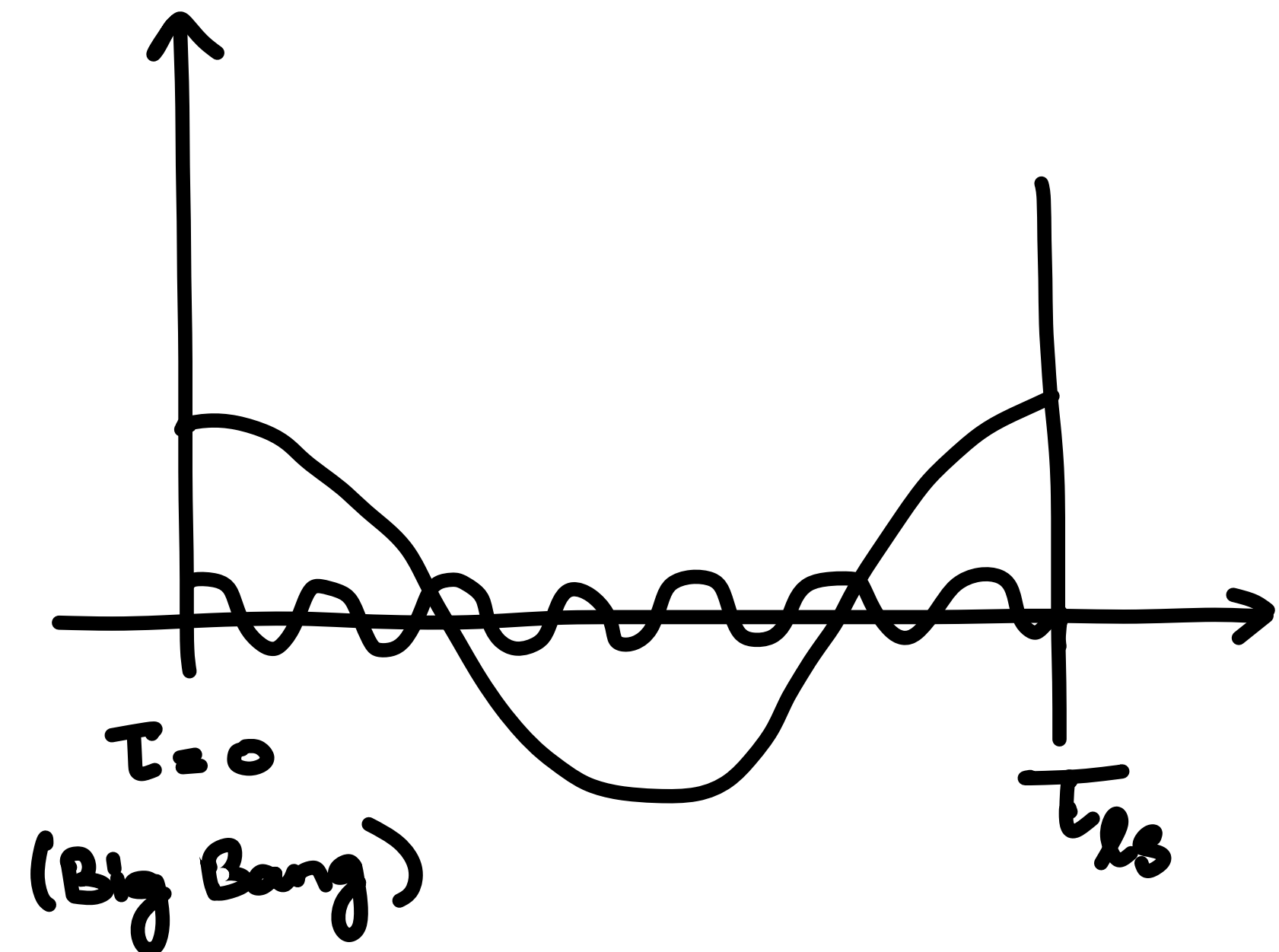
CMB Power spectrum: $P_\ell = \langle \Theta_\ell, \Theta_\ell \rangle$; $\left(\Theta_\ell = \frac{\delta T}{T} \right)$

Big Bang as a Mirror

Case1: Big Bang is a mirror (standing waves) Boyle & Turok 1803.08928

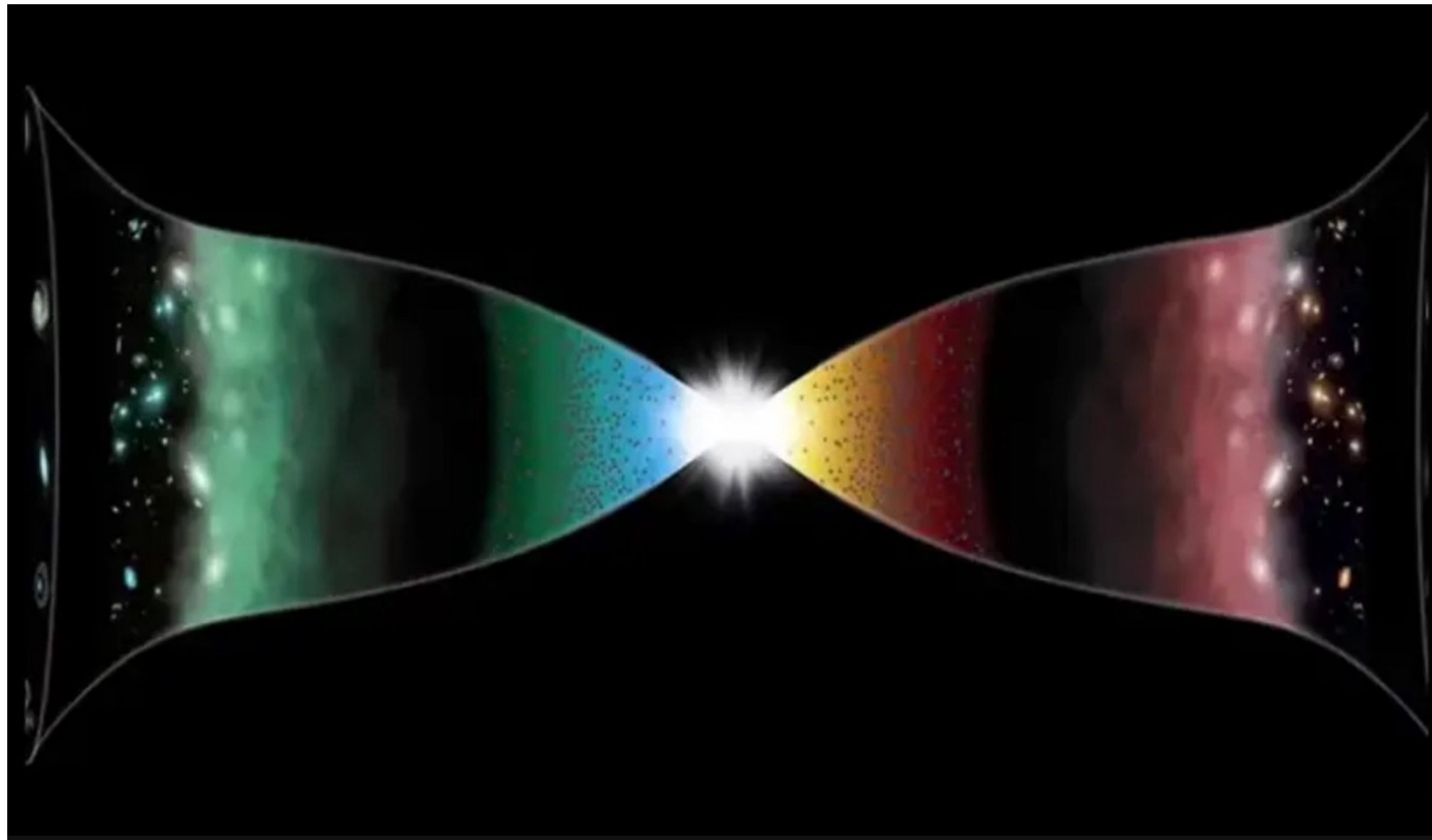


$$\left(\frac{\delta T}{T}\right)_k + \phi_k \approx A_k \cos\left(k \int_0^{\tau_k} c_s d\tau'\right) - \frac{3}{4} \frac{P_0}{P_k} \phi_k$$



Big Bang as a Mirror

$$ds^2 = a^2(\tau) [-d\tau^2 (1+\Phi) + 2\xi_i d\tau dx^i + dx^i dx^j (S_i - (1-\Phi)\delta_{ij} + h_{ij})]$$



- $a(-\tau) = -a(\tau)$
- $\Phi(\tau, \underline{x}) = \Phi(-\tau, \underline{x})$
- $\xi_i = 0$
- $h_{ij}(\tau, \underline{x}) = h_{ij}(-\tau, \underline{x})$

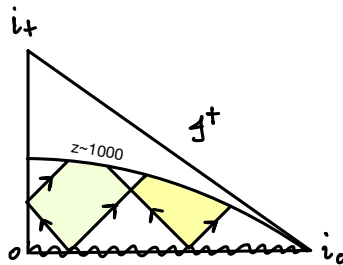
Problems with Extending Radiation Era to the Bang

- ▶ Horizon Problem
- ▶ Overproduction of dark matter/ thermalization of gravitons
- ▶ How to get scale-invariant power spectrum?

Content

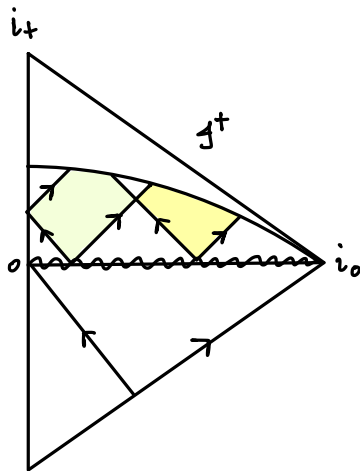
- ▶ Horizon problem
- ▶ Dark matter overproduction in Einstein gravity
- ▶ Review of quadratic gravity
- ▶ Dark matter thermalization, correction due to quadratic gravity
- ▶ Primordial perturbations and quadratic gravity

Horizon Problem



Radiation

Horizon Problem



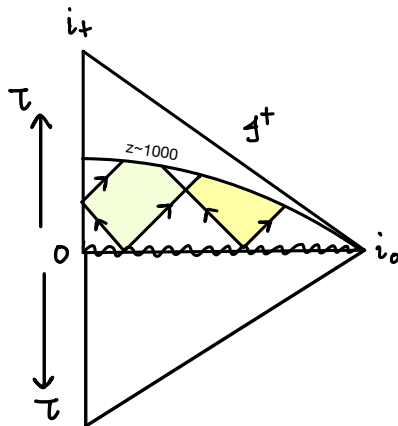
Inflation

Horizon Problem

- ▶ N. Turok and L. Boyle, Gravitational entropy and the flatness, homogeneity and isotropy puzzles, Phys. Lett. B 849, 138443 (2024), arXiv:2201.07279 [hep-th].

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The Right-Handed Neutrino Dark Matter

$$i\bar{l}_L\not{\partial}l_L + i\bar{e}_R\not{\partial}e_R + i\bar{\nu}_R\not{\partial}\nu_R - \bar{l}_LY_e^\dagger e_R h - \bar{l}_LY_\nu^\dagger \nu_R h^{(c)} - \frac{1}{2}M^\dagger\bar{\nu}_R^{(c)}\nu_R + h.c.$$

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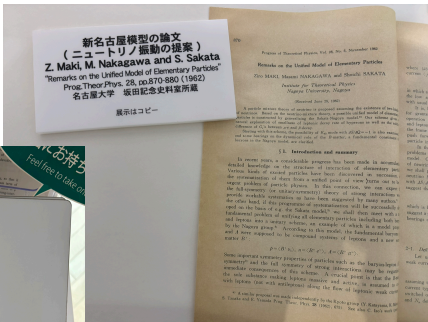
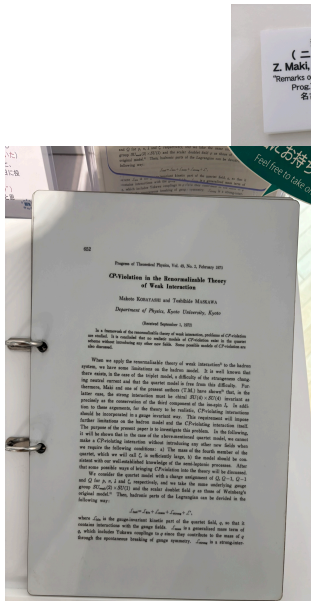
- ▶ Stabilize one of the right-handed neutrinos under a $\mathbb{Z}_2$ symmetry $e_R^{(1)} \rightarrow -e_R^{(1)}$.
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- ▶ The right-handed neutrino dark-matter can still have a Majorana mass, but the corresponding flavor of left-handed neutrino will be predicted to be massless in this theory.

The Legacy of Sakata and Nagoya



DM Production Through Cosmic Expansion

- ▶ If the dark-matter is strictly not interacting with the Standard Model, it will never freeze-in. How then is it produced in the early universe?

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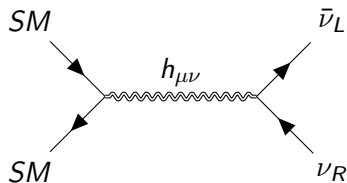
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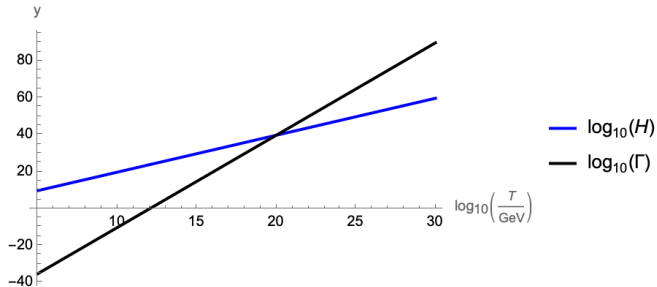
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- ▶ The expansion of the Universe itself produces it!
- ▶ Dark matter can also be produced because of QFT on curved space effects. A late time FRW observer will see a non-zero dark-matter number density corresponding to the vacuum state of an early observer.
- ▶ The dark-matter yield calculated through this process has been compared to the dark-matter yield observed today assuming no first order phase transition in the Universe. This predicts the mass of the right-handed neutrino dark matter to be $4.8 \times 10^8 \text{ GeV}$ (Boyle, Finn, Turok 1803.08930).

Thermalization of Dark Matter



Standard Model particles interacting with the Dark matter through the graviton channel.

Thermalization of Dark Matter (Einstein Gravity)



The transition rate for Dark Matter- Standard Model interaction shoots Hubble rate in Einstein gravity

$$\Gamma \sim 4g_* \frac{T^5}{M_{Pl}^4} > H = \left(\frac{\pi^2}{90}g_*\right)^{1/2} \frac{T^2}{M_{Pl}} \quad (2)$$

Quadratic Gravity

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- ▶ The quadratic in curvature terms have dimensionless couplings and are negligible in the infrared where Einstein gravity remains unchanged.
- ▶ The UV theory is renormalizable unlike Einstein gravity but comes with the baggage of ghost particles that have a negative norm.

Propagator for Gravity

- ▶ Let us consider linear fluctuations around a flat background $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, where $\eta_{\mu\nu} = (+, -, -, -)$
- ▶ We have the following orthogonal projectors:

$$\begin{aligned}\mathcal{P}_{\mu\nu\alpha\beta}^2 &= \frac{1}{2}(\theta_{\mu\beta}\theta_{\nu\alpha} + \theta_{\mu\alpha}\theta_{\nu\beta}) - \frac{1}{3}\theta_{\mu\nu}\theta_{\alpha\beta} \\ \mathcal{P}_{\mu\nu\alpha\beta}^1 &= \frac{1}{2}(\theta_{\mu\alpha}\omega_{\nu\beta} + \theta_{\nu\beta}\omega_{\mu\alpha} + \theta_{\mu\beta}\omega_{\nu\alpha} + \theta_{\nu\alpha}\omega_{\mu\beta}) \\ \mathcal{P}_{\mu\nu\alpha\beta}^0 &= \frac{1}{3}\theta_{\mu\nu}\theta_{\alpha\beta} \\ \tilde{\mathcal{P}}_{\mu\nu\alpha\beta}^0 &= \omega_{\mu\nu}\omega_{\alpha\beta}\end{aligned}\tag{4}$$

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- ▶ where

$$\theta_{\mu\nu} = \eta_{\mu\nu} - \frac{q_\mu q_\nu}{q^2}, \quad \omega_{\mu\nu} = \frac{q_\mu q_\nu}{q^2}\tag{5}$$

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- ▶ where the propagator in fourier space is given by:

$$i\mathcal{D}_{\mu\nu\alpha\beta}(q) = \frac{-2if_2^2}{q^2(q^2 - f_2^2 m^2)} \mathcal{P}_{\mu\nu\alpha\beta}^2 + \frac{if_0^2}{q^2(q^2 - f_0^2 m^2)} \mathcal{P}_{\mu\nu\alpha\beta}^0 \quad (7)$$

Propagator for Gravity

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- ▶ whereas the UV propagator is quartic instead of quadratic

$$i\mathcal{D}_{\mu\nu\alpha\beta}^{UV}(q) = \frac{-2if_2^2}{q^4} \mathcal{P}_{\mu\nu\alpha\beta}^2 + \frac{if_0^2}{q^4} \mathcal{P}_{\mu\nu\alpha\beta}^0 \quad (10)$$

Ghost in the Tensor Mode?

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- ▶ A mechanism to explain the correlation between superhorizon modes in the early (trans-Planckian) universe?

Asymptotically Free or Landau Pole?

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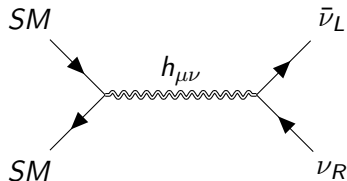
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- ▶ We will take the perspective that the theory is renormalizable and tachyon free but hits a Landau pole at extremely high energies just like in QED.

Thermalization of Dark Matter (Quadratic Corrections)

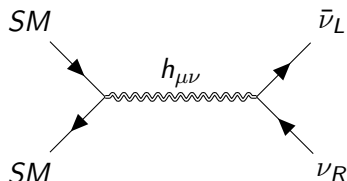


- The infrared graviton propagator is quadratic and gives a transition rate that grows as fifth power of the temperature.

$$i\mathcal{D}_{\mu\nu\alpha\beta}^{IR}(q) = \frac{i}{m^2 q^2} (2\mathcal{P}_{\mu\nu\alpha\beta}^2 - \mathcal{P}_{\mu\nu\alpha\beta}^0) \quad (11)$$

$$\Gamma \sim 4g_* \frac{T^5}{M_{Pl}^4} > H = \left(\frac{\pi^2}{90} g_* \right)^{1/2} \frac{T^2}{M_{Pl}} \quad (12)$$

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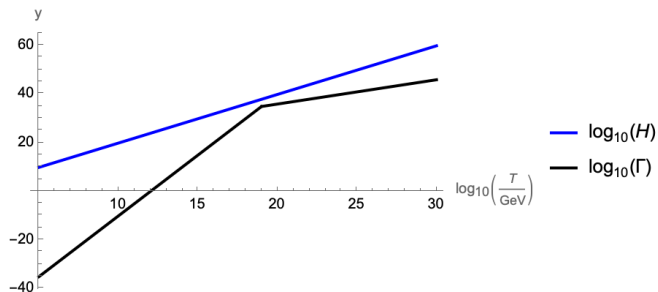


- The UV propagator is quartic which results in absence of scale and the transition rate now grows linearly with the temperature.

$$i\mathcal{D}_{\mu\nu\alpha\beta}^{IR}(q) = \frac{i}{q^4}(-2f_2^2\mathcal{P}_{\mu\nu\alpha\beta}^2 + f_0^2\mathcal{P}_{\mu\nu\alpha\beta}^0) \quad (13)$$

$$\Gamma \sim g_* f_2^4 T \quad (14)$$

Thermalization of Dark Matter



The transition rate is suppressed in the UV when quadratic curvature terms start dominating. $f_2 < \mathcal{O}(1)$ ensures $\Gamma < H$.

Dark Matter Yield Through Thermal Channel

- Stronger constraint coming from conservation of yield (assuming no first order phase transition)

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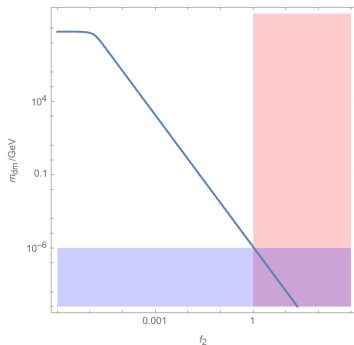
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- ▶ The integral above gives the following yield:

$$\frac{n_{dm}}{s} = \frac{N}{3.6g_*} = \left(\frac{1}{128 \times 3.6^2 \pi^3 g_*^3} \right)^{1/2} f_2^3 \quad (17)$$

Contour Plot of f_2 and Dark Matter Mass m_{dm}

Dark matter overproduction and unitarity violation Too warm to be dark matter



Constraint on the Weyl squared coupling f_2 and the dark matter m_{dm} coming from conservation of dark matter yield.

Constraints from the Tensor to Scalar Ratio

- ▶ The PLANCK + BICEP constraint on the tensor to scalar ratio (r) also constraints the Weyl squared coupling:

$$r = \frac{f_2^2}{A} < 0.036 \Rightarrow f_2 < 8.69 \times 10^{-6} \quad (18)$$

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- ▶ For this very small value of the coupling f_2 , the mass of the dark matter right-handed neutrino is predicted to be $4.8 \times 10^8 \text{ GeV}$.

Primordial Power Spectrum

- ▶ $R^{(3)} = -4\nabla^2\mathcal{R}$ is the Ricci scalar for a constant time-slice and the comoving curvature perturbation $\mathcal{R}$ is a quantity conserved in time.
- ▶ The 2-point correlation between thermal fluctuations in the CMB is related to the 2-point correlation function of the comoving curvature perturbation $\mathcal{R}$.
- ▶ From observations we know that on super horizon scale, the 2-point correlation function of $\mathcal{R}$ is almost scale invariant.

$$\langle \mathcal{R}(\vec{x}), \mathcal{R}(\vec{y}) \rangle = \int d\ln(k) \mathcal{P}_k e^{i\vec{k} \cdot (\vec{x} - \vec{y})} \quad (19)$$

$$\mathcal{P}_k = A \left(\frac{k}{k_*} \right)^{n_s - 1}, A \sim 2 \times 10^{-9}, n_s \sim 0.97 \quad (20)$$

Primordial Power Spectrum

- ▶ One can show that the comoving curvature perturbation can be expressed in terms of the conformal perturbation of the metric in the long-wavelength approximation.

$$g_{\mu\nu} = a^2(\tau)(-d\tau^2 + d\vec{r}^2) \rightarrow e^{2\sigma} g_{\mu\nu} \approx (1 + 2\sigma(\tau, \vec{r}))g_{\mu\nu} \quad (21)$$

- ▶ The comoving curvature perturbation is given by:

$$\mathcal{R} = \sigma - \frac{2(\sigma - H^{-1}\partial_t\sigma)}{3(1+\omega)} \quad (22)$$

- ▶ In the long wavelength approximation $\partial_t\sigma \approx 0$ and in radiation era ($\omega = 1/3$) we get:

$$\mathcal{R} = \sigma/2 \quad (23)$$

Primordial Power Spectrum

- ▶ In flat background, we have seen that the propagator for the conformal scalar mode (or in other words the 2-point correlation function) is given by:

$$\langle \sigma(x^\mu), \sigma(y^\mu) \rangle = \int d^4k \frac{if_0^2 e^{ik \cdot (x-y)}}{k^2(k^2 - f_0^2 m^2)} \quad (24)$$

- ▶ In the UV it becomes scale-invariant hinting that the scalar amplitude is given in terms of the coupling f_0

$$f_0^2 = A \sim 2 \times 10^{-9} \quad (25)$$

- ▶ The spectral tilt can then be obtained in terms of the beta function of f_0 .

$$n_s - 1 = \frac{d \ln(\mathcal{P}_k)}{d \ln k} = \frac{2}{f_0} \beta_{f_0} \quad (26)$$

Primordial Power Spectrum

- ▶ The spectral tilt can then be obtained in terms of the beta function of f_0 .

$$n_s - 1 = \frac{d \ln(\mathcal{P}_k)}{d \ln k} = \frac{2}{f_0} \beta_{f_0} \quad (27)$$

$$= \frac{1}{(4\pi)^2} \left[\frac{5}{3} \frac{f_2^2}{f_0^2} + 5 f_2^2 \frac{5}{6} f_0^2 + \frac{f_0^2}{12} (1 + 6\xi) \right] \quad (28)$$

- ▶ Fine tuning the non-minimal coupling of Higgs $\xi \sim 10^8$ gives us the correct tilt but results in a huge α .
- ▶ Other possibilities under consideration.

Summary and Future Work

- ▶ We see that it is consistent to extend radiation era all the way to the Big Bang without overproducing dark matter.
- ▶ To do so we require higher derivative corrections to Einstein Gravity.
- ▶ Weyl squared coupling is constrained to be very small $\lesssim 10^{-6}$
- ▶ Pure quadratic gravity (without EH term) can explain the scale invariance of the primordial power spectrum even without Starobinsky's inflationary solution.
- ▶ The scalar amplitude will constraint the R^2 coupling, while the spectral-tilt might put a strong constraint on the non-minimal coupling of scalar fields.